

26
Rafale-Ms to be acquired by the Navy

₹65,920
CRORE
Total Cost

1,389
KMPH
Max speed

KAMOV KA-28 12
Anti Submarine Warfare helicopters

ALH DHRUV 18
Multi-role and multi-mission indigenous Advanced Light helicopter



MH60R HELICOPTERS 5
Anti-Submarine Warfare, surveillance, and search and rescue duties (five delivered out of a total deal for 24)

CHETAK
Search and rescue multi-purpose helicopters

● Numbers in service



electronic scanning radar, Front Sector Optronics (FSO) that allows targets to be detected and tracked and SPECTRA internal 'electronic warfare' (EW) systems for long-range detection and identification of hostile radars, missiles and lasers. In addition, Rafale's unique data fusion capability yields a simplified and consolidated picture to the pilot. The Rafale-M is equipped with the Safran Group's reinforced landing gear as well as tailhooks that attach themselves to cables strung across the carrier deck, enabling the aircraft to be brought to a stop.

Former navy chief Admiral Arun Prakash, a naval pilot himself, endorses the navy's choice, adding that Rafale-M is currently the most advanced carrier-borne jet in its class. "It has got good sensors, can carry a variety of weapons and has good endurance, which brings tremendous enhancement of capability. Carrier-based fighters can go 300-400 miles to strike enemy territory or targets, affording an enhanced strike range. It provides air defence against enemy attacks and can be the navy's eyes and ears in the sky over the sea," Adml Prakash adds.

Defence officials say that since the IAF is flying 36 Rafale jets, which shares over 80 per cent of its components with the Rafale-M, the navy's deal will bring commonality as well as savings on training and maintenance. According to Dassault, after the Navy gets its Rafale-Ms, it would make India the first country after France to operate both versions of the aircraft.

The Rafale-M can carry up to 9.5 tonnes of weapons, fuel tanks and pods on 13 external hardpoints, and is capable of low-level, high-speed penetrations to evade enemy radar. It can be equipped with weapons like the METEOR very long-range air-to-air missile, MICA air-to-air "Beyond Visual Range" (BVR) interception, combat and self-defence missiles, in their IR (infrared) heat-seeking (detecting hot engine plumes of targets) and active radar homing (using inbuilt radars to detect targets) versions. It can also use the HAMMER (Highly Agile and Manoeuvrable Munition Extended Range) modular, rocket-boosted air-to-ground precision-guided weapon series, SCALP long-range stand-off missile, AM39 EXOCET anti-ship missile, laser-guided bombs with different warheads and seekers, and the 2,500 rounds/minute NEXTER 30M791 30 mm internal cannon. The website of Dassault Aviation says the Rafale-M is capable of several complex actions during the same sortie—termed 'omnirole' capability—like firing air-to-air missiles during a low-altitude penetration phase.

Built to withstand the challenges of saltwater environments, naval jets incorporate corrosion-resistant materials and coatings. Accordingly, Rafale-M has a robust airframe and undercarriage made of advanced composite materials and corrosion-resistant alloys to provide protection against rust, and to ensure perfect aircraft performance even in severe environments. While naval fighter jets generally have shorter ranges, they boast in-flight refuelling capabilities, ensuring extended operational endurance during critical missions. Rafale-M is no exception.

According to Dassault, the French Navy Rafale-Ms operating from the aircraft carrier *Charles De Gaulle* served with efficiency in Syria in 2015. In 2011, Rafales and Rafale-Ms also took part in operations over Libya.

The effective use of these latest jets from the deck of India's new, indigenous aircraft carrier can yield significant results. Carriers are ideal for sea control and power projection. In military parlance, sea control is exercised using a combination of warships, fixed-wing aircraft, helicopters and amphibious platforms. INS *Vikrant* and INS *Vikramaditya* are crucial for operations in the Indian Ocean Region, where, too, India faces the spreading tentacles of an aggressive China.

"Fighter jets for an aircraft carrier are like the eyes, ears and soul of the fleet. Without fighter jets, the aircraft carrier fleet is helpless," says Adml Prakash. The 26 Rafale-Ms are expected to play that role to perfection and help further India's maritime objectives. ■